RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 2C: Vectors III – Equations of Planes

1 The line l passes through the points A and B with coordinates $(1,2,4)$ and $(-2,3,1)$ respectively. The plane p has equation $3x-y+2z=17$. Find

(i) the coordinates of the point of intersection of l and p, [5]

(ii) the acute angle between l and p, [3]

(iii) the perpendicular distance from A to p. [3]

2 The equations of a plane π_1 and a line l are shown below:

$$\pi_1 : \mathbf{r}.\begin{pmatrix}2\\1\\-2\end{pmatrix}=6$$

$$\frac{x-1}{3}=\frac{y+4}{2}=z-5$$

The point A has position vector $3\mathbf{i}-\mathbf{j}+4\mathbf{k}$.

(i) Find the distance between point A and the plane π_1. [2]

(ii) B is another point such that $\overrightarrow{AB}=-5\mathbf{j}-2\mathbf{k}$. Find the length of projection of $\overrightarrow{AB}$ onto the plane π_1. [2]

(iii) Using your answers in **(i)** and **(ii)**, find the area of triangle ABC, where C is the reflection of A in the plane π_1. [2]

(iv) Find the equation of the plane π_2 which contains the line l and the origin. Hence, find the line of intersection between the planes, π_1 and π_2. [4]

3 The line ℓ whose vector equation is $\mathbf{r}=5\mathbf{i}+2\mathbf{j}-3\mathbf{k}+\alpha(7\mathbf{i}-2\mathbf{j}-3\mathbf{k})$ passes through the point A with position vector $5\mathbf{i}+2\mathbf{j}-3\mathbf{k}$. The plane π whose vector equation is

$$\mathbf{r}=-5\mathbf{i}+2\mathbf{j}+2\mathbf{k}+\beta(7\mathbf{i}-2\mathbf{j}-3\mathbf{k})+\gamma(9\mathbf{i}-\mathbf{j}+\mathbf{k})$$

contains the point B with position vector $-5\mathbf{i}+2\mathbf{j}+2\mathbf{k}$.

(i) Find the position vector of the point P on the line segment AB such that $AP:PB=4:1$. [2]

(ii) The plane π_1 contains the line ℓ and the point P.

Write down a vector equation of the line of intersection of π and π_1. [2]

Find the vector equation of π_1 and the angle between π and π_1. [5]

Hence find the ratio of the perpendicular distances from P to the line ℓ and from P to the plane π. [2]

4 A plane π_1 has equation $\mathbf{r}\cdot\begin{pmatrix}2\\-1\\3\end{pmatrix}=9$ and a line ℓ_1 has equation $\mathbf{r}=\begin{pmatrix}3\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\-2\end{pmatrix}$.

(i) Find the coordinates of P, the point of intersection of ℓ_1 and π_1. [3]

Hence, or otherwise, find the shortest distance from point A (3, 0, 0) to π_1. [2]

The equation of a plane π_2 is given as

$$\pi_2 : \quad \mathbf{r}=\begin{pmatrix}1\\1\\1\end{pmatrix}+s\begin{pmatrix}2\\0\\-3\end{pmatrix}+t\begin{pmatrix}2\\2\\1\end{pmatrix}$$

(ii) Find the equation of plane π_2 in the form $\mathbf{r}\cdot\mathbf{n}=d$. Explain why the planes π_1 and π_2 intersect. [4]

5 The position vectors of the points A, B, C and D are given as $\mathbf{i}+3\mathbf{j}$, $2\mathbf{j}+4\mathbf{k}$, $\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $4\mathbf{i}+5\mathbf{k}$ respectively.

(i) Find the vector equation of plane π in the form $\mathbf{r}\cdot\mathbf{n}=p$, that contains the points A, B and C. [3]

(ii) Find the foot of the perpendicular of the point D to the plane π. Hence find the shortest distance from the point D to the plane π. [4]

A line l parallel to the vector $\mathbf{j}+\mathbf{k}$ passes through point D and it meets the plane π at the point N.

Find the position vector of the point N and hence find the vector equation of the reflection of line l about the plane π. [5]

6 **(i)** Interpret geometrically the vector equation $\mathbf{r}=\mathbf{a}+t\,\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors and t is a parameter. [2]

(ii) Interpret geometrically the vector equation $\mathbf{r}.\mathbf{n}=d$, where $\mathbf{n}$ is a constant unit vector and d is a constant scalar, stating what d represents. [3]

(iii) Given that $\mathbf{b}.\mathbf{n}\neq 0$, solve the equations $\mathbf{r}=\mathbf{a}+t\,\mathbf{b}$ and $\mathbf{r}.\mathbf{n}=d$ to find $\mathbf{r}$ in terms of $\mathbf{a},\mathbf{b},\mathbf{n}$ and d. Interpret the solution geometrically. [3]

7 NJCMidYear9740/2007/1/Q12 modified

The plane Π_1 has equation $\mathbf{r} \cdot (-\mathbf{i} + 2\mathbf{k}) = -4$ and the points A and P have position vectors $4\mathbf{i}$ and $\mathbf{i} + \alpha\mathbf{j} + \mathbf{k}$ respectively, where $\alpha \in \mathbb{R}$.

(i) Show that A lies on Π_1 but P does not. [1]

(ii) Find, in terms of α, the position vector of N, the foot of perpendicular of P on Π_1. [3]

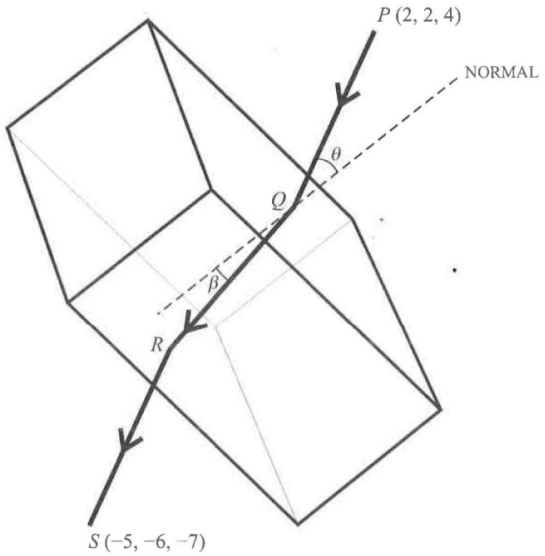
The plane Π_2 contains the points A, P and N.

(iii) Show that the equation of Π_2 is $\mathbf{r} \cdot (2\alpha\mathbf{i} + 5\mathbf{j} + \alpha\mathbf{k}) = 8\alpha$ and write down the equation of l, the line of intersection of Π_1 and Π_2. [3]

The plane Π_3 has equation $\mathbf{r} \cdot (\mathbf{i} + \mathbf{j} + 2\mathbf{k}) = 4$.

(iv) By considering l, or otherwise, find the value of α for which the three planes intersect in a line. [2]

8 9758/2019/01/Q12



A ray of light passes from air into a material made into a rectangular prism. The ray of light is sent in direction $\begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix}$ from a light source at the point P with coordinates $(2, 2, 4)$. The prism is placed so that the ray of light passes through the prism, entering at the point Q and emerging at the point R and is picked up by a sensor at point S with coordinates $(-5, -6, -7)$. The acute angle between PQ and the normal to the top of the prism at Q is θ and the acute angle between QR and the same normal is β (see diagram).

It is given that the top of the prism is a part of the plane $x+y+z=1$, and that the base of the prism is a part of the plane $x+y+z=-9$. It is also given that the ray of light along PQ is parallel to the ray of light along RS so that P, Q, R and S lie in the same plane.

(i) Find the exact coordinates of Q and R. [5]

(ii) Find the values of $\cos\theta$ and $\cos\beta$. [3]

(iii) Find the thickness of the prism measured in the direction of the normal at Q. [3]

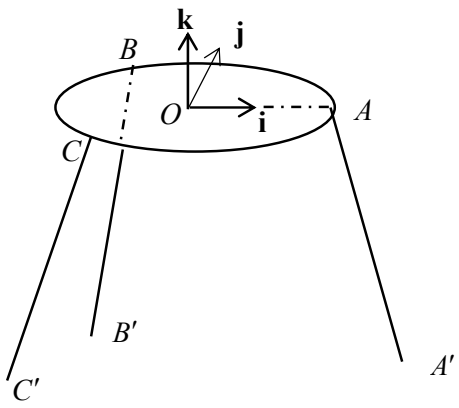
Snell's law states that $\sin\theta = k\sin\beta$, where k is a constant called the refractive index.

(iv) Find k for the material of this prism. [1]

(v) What can be said about the value of k for a material for which $\beta > \theta$? [1]

9 9205/1999/02/Q15

The diagram shows a circular table, with centre O, and three legs AA', BB' and CC', attached at the points A, B and C to the table. The lengths of the legs are adjusted so that, although the table is standing on a sloping floor, the plane ABC is horizontal. Perpendicular vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are defined with $\mathbf{i}$ along $\overrightarrow{OA}$, and $\mathbf{k}$ vertically upwards.



The vectors $\overrightarrow{AA'}$, $\overrightarrow{BB'}$ and $\overrightarrow{CC'}$ are parallel to $5\mathbf{i}-12\mathbf{k}$, $-3\mathbf{i}+4\mathbf{j}-12\mathbf{k}$ and $-3\mathbf{i}-4\mathbf{j}-12\mathbf{k}$ respectively.

(i) Show that all three legs are inclined at the same angle to the vertical. [3]

(ii) The plane $A'B'C'$ has equation $x-10z=130$. Find the inclination of this plane to the horizontal, and the perpendicular distance from O to this plane. [4]

(iii) Referred to O as the origin, the position vectors of A, B and C are $5\mathbf{i}$, $-3\mathbf{i}+4\mathbf{j}$ and $-3\mathbf{i}-4\mathbf{j}$ respectively. Find the lengths of the legs AA', BB' and CC'. [7]

10 9205/1997/2/Q15

The plane π passes through the origin O and is perpendicular to $2\mathbf{i}+2\mathbf{j}+\mathbf{k}$. The line L has Cartesian equation $\frac{x-4}{1}=\frac{y-8}{3}=\frac{z-4}{2}$. The points A and B have position vectors $\mathbf{i}+2\mathbf{j}-3\mathbf{k}$ and $3\mathbf{i}+5\mathbf{j}+2\mathbf{k}$ respectively.

(i) State a Cartesian equation for π. [1]

(ii) The point C on L has position vector $\mathbf{c}$, where $\mathbf{c}\cdot\mathbf{k}=0$. Find $\mathbf{c}$, and hence state an alternative equation for L of the form $\frac{x-p}{l}=\frac{y-q}{m}=\frac{z}{n}$, giving numerical values for p, q, l, m, and n. [4]

(iii) Find the exact value of the length of perpendicular from A to L. [2]

(iv) Find, in either order, the exact value of the length of projection of $\overrightarrow{AB}$ onto

(a) the normal to π, [2]

(b) the plane π. [2]

THE END